SCHEME-THEORETIC SUPPORT OF THE KERNEL OF THE PULLBACK OF AN ALMOST EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-local-cohomology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: the chapter carries no chapter-local bibliography and neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The notion of an (A, n, c)-module, and the closed subschemes Y_c of the affine blowup algebra used to measure scheme-theoretic support. Set up in the author's section on a bit of uniformity, where the counted statement occurs. Fixes the hypothesis on M and the conclusion about where the kernel is supported.
- The derived-functor description of Tor over a quotient ring and the effect of multiplying by an element of a power of the ideal. Used inside the counted proof at source lines 5116-5120. Turns the annihilation of Tor by a power of the ideal into the scheme-theoretic support statement.

Lemma 0.1. *Let $0 \to K \to L \to M \to 0$ be a short exact sequence of A-modules such that K and L are annihilated by I^n and M is an (A, n, c)-module. Then the kernel of $p^*K \to p^*L$ is scheme theoretically supported on Y_c.*

Proof. Let $\mathrm{Spec}(B) \subset X$ be an affine open. The restriction of the exact sequence over $\mathrm{Spec}(B)$ corresponds to the sequence of B-modules

$$K \otimes_A B \to L \otimes_A B \to M \otimes_A B \to 0$$

which is isomorphismic to the sequence

$$K \otimes_{A/I^n} B/I^nB \to L \otimes_{A/I^n} B/I^nB \to M \otimes_{A/I^n} B/I^nB \to 0$$

Hence the kernel of the first map is the image of the module $\mathrm{Tor}_1^{A/I^n}(M, B/I^nB)$. Recall that the exceptional divisor Y is cut out by $I\mathcal{O}_X$. Hence it suffices to show that $\mathrm{Tor}_1^{A/I^n}(M, B/I^nB)$ is annihilated by I^c. Since multiplication by $a \in I^c$ on M factors through a finite free A/I^n-module, this is clear. □

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.